Architecture

UNZA Hostel Maintenance Request and Tracking System

University of Zambia — Final Year Project

Architecture

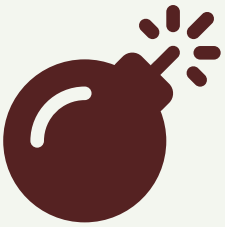
Hostel Maintenance Request and Tracking System — University of Zambia. A Progressive Web App (PWA) where students report hostel faults, administrators triage and assign them, and technicians resolve them, with the whole lifecycle tracked and analysed.

This document explains *how the system is built and why*. Diagrams use Mermaid, which GitHub renders automatically.

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1. The big picture

Client–server architecture; the server is internally a three-layer architecture; the client is a PWA.



Syntax error in text

mermaid version 11.17.2

```
(pages + components)"]
    SW["Service Worker
  (caching, push)"]
  Cache["In-memory API cache
+ IndexedDB draft queue"]
          end

subgraph Server["SERVER TIER — Node.js / Express"]
              direction TB
            P["Presentation layer
          routes + middleware
    (validate, authenticate, authorise)"]
            B["Business logic layer
              services
  (lifecycle rules, SLA, ranking, triage)"]
            D["Data access layer
            repositories
  (only place Supabase is imported)"]
            Jobs["Background jobs
   (overdue monitor, weekly email)"]
            P --> B --> D
              Jobs --> B
                 end

subgraph Data["DATA TIER — Supabase"]
            PG[("PostgreSQL
      + Row Level Security")]
          Auth["Auth (JWT)"]
            Store["Storage
            (photos)"]
              end

    UI -- "HTTPS, REST/JSON
  + Server-Sent Events" --> P
      D -- "Supabase SDK
    (service role)" --> PG
            D --> Auth
            D --> Store
   Server -. "Web Push" .-> SW
```

Two rules that are enforced everywhere

1. **The client never talks to Supabase.** There are no Supabase keys or SDK in the front end. Every read and write goes through the Express REST API. This is what keeps business rules and access control in one place where they can be enforced and tested.
2. **The server's layers only call downward.**
   - *Routes* validate input, authenticate, authorise, call one service, format the response. No business logic, no database access.
   - *Services* hold the business rules. They contain no `req` / `res`, so they are framework-agnostic and unit-testable in isolation.
   - *Repositories* are the only files that import the Supabase client. They query and map rows; no business logic.

> Why this matters: the layer boundary is what made every feature in this project small to add and safe to test. For example the SLA rule (`server/src/utils/sla.js`) is a pure function used by the API, the overdue monitor and the weekly email — one definition, three uses, six unit tests.

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2. Technology choices

Concern	Choice	Why
UI	React (functional components + hooks)	Component model fits the three role-based UIs; one codebase for phone and desktop
Styling	Plain CSS, one file per component, design tokens as CSS variables	No build-time dependency; tokens make theming (dark mode) a matter of overriding variables; class names are prefixed per component to avoid global clashes
Server	Node.js + Express	Same language end-to-end; Express middleware maps directly onto the presentation layer
Database	Supabase (PostgreSQL)	Relational integrity (foreign keys), Row Level Security, managed Auth and Storage
Auth	Supabase Auth (JWT), verified on the server	Password handling is not hand-rolled
PWA	Service worker + Web App Manifest	Installable, resilient to poor connectivity, receives push notifications
Charts	Recharts (the only charting library)	Animated, accessible charts; loaded only for administrators
Push	Web Push (VAPID) via `web-push`	Standards-based, no third-party push service
Email	SMTP via `nodemailer`	Works with any provider (Gmail app password is enough)

Concern	Choice	Why
AI (optional)	Anthropic Claude via the official SDK	Category/priority suggestions; the app works without it

Everything is JavaScript, on purpose: one language for students of the codebase.

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3. Folder structure

```
client/                    React PWA
  public/                  manifest.json, service-worker.js, icons
  src/
    pages/                 one folder per screen (.jsx + .css)
    components/            shared UI (StatCard, SlaBadge, charts, ...)
    services/              API wrappers, cache, offline draft queue, push, theme
    hooks/                 auth, live updates
    styles/                variables.css (tokens) + theme.css (dark theme)
    utils/                 formatters, image compression
server/
  server.js                Express app + scheduler start
  sql/                     numbered migrations 001 ... 012
  src/
    routes/                presentation layer
    middleware/            authenticate, requireRole, errorHandler
    services/              business logic layer (+ unit tests)
    repositories/          data access layer (only place Supabase is imported)
    jobs/                  background scheduler
    config/                Supabase, web-push, mail, Anthropic clients
    utils/                 pure helpers (SLA rules, labels)
docs/                      this document, viva guide, screenshots
```

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4. Data model

```
erDiagram
    users ||--o{ maintenance_requests : "reports (student_id)"
    categories ||--o{ maintenance_requests : classifies
    maintenance_requests ||--o{ technician_assignments : "assigned via"
    users ||--o{ technician_assignments : "works (technician_id)"
    maintenance_requests ||--o{ status_history : "logs every transition"
    maintenance_requests ||--o| ratings : "rated once"
    maintenance_requests ||--o{ request_upvotes : "backed by"
    users ||--o{ request_upvotes : casts
    maintenance_requests ||--o| resolution_proofs : "proof of work"
    users ||--o{ push_subscriptions : "devices"

    users {
        uuid user_id PK
        text name
        text email UK
        text id_number UK
        enum role "student | technician | administrator"
        enum availability_status
    }
    maintenance_requests {
        int request_id PK
        uuid student_id FK
        int category_id FK
        text description
        text photo_url
        enum status "open..closed"
        enum priority "low..critical"
        text hostel
        timestamptz created_at
        timestamptz overdue_notified_at
    }
    status_history {
        int history_id PK
        int request_id FK
        enum previous_status
        enum new_status
        timestamptz changed_at
    }
    request_upvotes {
        int request_id PK
        uuid user_id PK
    }
    resolution_proofs {
```

```
        int request_id PK
        text photo_url
        text note
        uuid submitted_by FK
        }
```

Design points worth defending:

- **Integrity lives in the database, not only in code.** Foreign keys everywhere; `request_upvotes` has a *composite primary key* `(request_id, user_id)` so a double vote is impossible even if application code were bypassed; `ratings` has a unique `request_id` (one rating per request).
- **Row Level Security is on for every table** as a second line of defence (students see only their own rows, technicians only assigned work, administrators broader access). The server uses the service role, so RLS matters most if a key ever leaked or a client were ever pointed at the database directly.
- **Every status change is recorded** in `status_history`. This is what feeds the analytics (average resolution time, throughput) and the timeline users see.
- **Additive migrations.** New features add tables rather than altering hot ones (e.g. `resolution_proofs` is its own table), so the app keeps working before a migration has been applied.

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5. The ticket lifecycle (a business rule, not a UI rule)

```
stateDiagram-v2
    [*] --> Open: student submits
    Open --> Assigned: administrator assigns a technician
    Assigned --> InProgress: technician starts
    InProgress --> Resolved: technician resolves (photo optional)
    Resolved --> Closed: closed out
    Closed --> [*]
```

- Strictly linear, one step at a time; skipping (`Open → Closed`) or going backward is rejected **by the service layer** with a clear error (`requests.service.js → isValidTransition`).
- `Assigned` can only be reached through the assign action, which creates the `technician_assignments` row in the same operation — status and assignment cannot drift apart.
- A technician can only advance requests **assigned to them**; students only see their own; administrators see all. Checked on the server on every request.
- When resolving, a technician can attach a **photo of the finished work** and a note (optional; `REQUIRE_RESOLUTION_PROOF=true` makes the photo mandatory for technicians). If a photo is sent it is validated and stored *before* the status changes, so a failed upload never leaves a request "resolved" with lost proof.

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6. Authentication and access control

ID-based login, resolved on the server. Students log in with their student/staff ID, not their email, but Supabase Auth is email-based underneath:

```
sequenceDiagram
participant C as Client
participant R as Route (auth.routes)
participant S as auth.service
participant DB as users table
participant SA as Supabase Auth
C->>R: POST /auth/login { id_number, password }
R->>S: login(...)
S->>DB: find email for id_number
S->>SA: signInWithPassword(email, password)
SA-->>S: session (JWT)
S-->>C: session + profile
```

Every protected route then runs `authenticate` (verifies the JWT and loads the profile) and `requireRole(...)`. **The UI hiding a button is convenience; the server check is the security.** Ownership rules (a student may read only their own request, a technician only assigned ones) live in the services.

Profile lookups are cached for 30 seconds (cleared on profile edits) to remove two network round trips from most requests.

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7. Real-time behaviour

Feature	Mechanism
Open pages update when a request changes	**Server-Sent Events** (`GET /api/events`). The server pushes only a request id + message; the client refetches through the normal permission-checked API. SSE was chosen over WebSockets/Supabase Realtime because updates only flow server → client and the architecture forbids the client talking to Supabase.
Notifications when the app is closed	**Web Push** (VAPID). Sent on new requests, assignments, status changes and overdue alerts. Logging out removes the device's subscription so a shared phone never receives someone else's alerts.
Overdue alerts	A background monitor checks every hour; a request past its SLA is alerted once, then reminded while still overdue (every 4 h for critical/high, daily otherwise — `overdue_notified_at` holds the last alert time) — to administrators + the assigned technician.
Weekly summary	Monday 06:00 UTC (08:00 Lusaka) email to administrators; last-run stored in `job_runs` so a restart cannot cause a duplicate.

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8. Offline tolerance and performance (the PWA layer)

- **Installable**: web manifest with PNG + maskable icons, standalone display, shortcuts (*Report an issue*, *My requests*). Chrome's installability check reports no errors.
- **Service worker strategies**: hashed build assets are cache-first (instant, no network); page loads are network-first with a 2.5 s timeout falling back to the cached shell; everything else network-first.
- **Offline drafting**: a student can write a request with no connectivity; it is stored in IndexedDB (`draftQueue.js`) and submitted automatically when the connection returns (`draftSync.js`). Implemented in the page rather than Background Sync, because Background Sync is not available on iOS/Firefox.
- **Stale-while-revalidate data cache** (`apiCache.js`): pages render instantly from the last data they saw and refresh in the background; the cache is cleared on any write and on login/logout so stale or another user's data never shows.
- **Code splitting**: every page is its own chunk. The entry bundle is ≈ 248 kB (79 kB gzipped); the 355 kB chart library is downloaded only by administrators. Remaining pages are prefetched during idle time.
- **gzip** on API responses; photos are compressed in the browser (max 1600 px, JPEG) before upload, turning 4–10 MB phone photos into a few hundred kB.

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9. Feature modules (beyond the core lifecycle)

Feature	Where it lives	Business rule
Upvotes ("Me too")	`upvotes.service.js`	Can't upvote your own or a resolved request; idempotent; other students' names are never exposed
AI triage	`triage.service.js`	Claude (schema-constrained JSON) with a keyword-rules fallback; a **safety floor** forces hazards (fire, sparking, flooding) to *critical* regardless of the model; user text is treated as untrusted; per-user rate limit
SLA timers	`utils/sla.js`	critical 4 h · high 24 h · medium 72 h · low 1 week; *at risk* in the last 25 %; stops when resolved
Smart assignment	`users.service.js → rankTechnicians`	Available before on-job before off-duty; then fewest active tasks; off-duty is never recommended
Proof of work	`requests.service.js`	Technicians can attach a photo (and note) when resolving; it is optional by default and can be made mandatory with `REQUIRE_RESOLUTION_PROOF=true`
Hostel heatmap	`HostelHeatmap.jsx`	Hostel × category counts; answers "where should maintenance go this week?"
Dark mode	`styles/theme.css`	Overrides colour tokens only, so every component switches automatically

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10. API surface

All routes are under `/api`. "Scoped" means the service checks ownership as well as role.

Area	Endpoint	Who
Auth	`POST /auth/register`	public (students only)
	`POST /auth/login`	public
	`GET /auth/me`	any signed-in user
Requests	`POST /requests`	student
	`GET /requests/mine`	student
	`GET /requests/community`	student
	`POST /requests/triage`	student
	`POST /requests/:id/upvote` and `DELETE /requests/:id/upvote`	student
	`POST /requests/:id/rating`	student
	`GET /requests`	administrator
	`POST /requests/:id/assign`	administrator
	`PATCH /requests/:id/priority`	administrator
	`GET /requests/assigned-to-me`	technician
	`GET /requests/:id`	scoped (owner / assigned / admin)
	`PATCH /requests/:id/status`	scoped (assigned technician / admin)
Users	`GET /users`, `POST /users`	administrator
	`GET /users/technicians`	administrator
	`GET /users/me`, `PATCH /users/me`	any signed-in user
	`PATCH /users/me/availability`	technician
Reports	`GET /reports/analytics`	administrator
	`POST /reports/digest/send`	administrator
Live and push	`GET /events` (Server-Sent Events)	any signed-in user
	`GET /push/public-key`	public
	`POST /push/subscribe`, `POST /push/unsubscribe`	any signed-in user
Misc	`GET /categories`, `GET /activity/recent`	any signed-in user
	`GET /health`	public

11. Security summary

- HTTPS in deployment; JWT verified on every protected route; passwords never handled by the app (Supabase Auth).
- Server-side validation of every input, regardless of client validation.
- Role checks (`requireRole`) **and** ownership checks in services; RLS as a second layer.
- No secrets in source control: `.env` is git-ignored, `.env.example` documents every variable. No Supabase key exists in the front end.
- Uploads: type allow-list (JPEG/PNG/WebP), 5 MB cap, stored server-side.
- Email templates HTML-escape user-controlled text; AI prompts treat descriptions as untrusted data; the AI endpoint is rate-limited.
- Push subscriptions are per user and removed on logout.

12. Testing

- **158 automated tests** (Vitest): **110 unit tests** covering the business rules with repositories mocked, and **48 integration tests** that drive the real Express app over HTTP (routes + middleware + services, database faked) — see TESTING.md. The unit tests cover: lifecycle transitions and role restrictions, proof-of-work ordering, SLA states, overdue detection, technician ranking, upvote rules and feed ranking, AI-triage safety floor and rate limit, push delivery and cleanup, live-event routing, weekly-email statistics/schedule/escaping, analytics maths, and the auth/role middleware.
- **Browser verification** (Playwright, headless Chromium) of the front-end flows at phone and desktop widths, including the offline path, live updates arriving on an open page, the technician proof flow and dark mode.
- **Manual UAT** of the student → administrator → technician flow.
- **Continuous integration** (GitHub Actions, `.github/workflows/ci.yml`): every push and pull request runs the unit tests, the integration tests, lint and a production build; no secrets are needed.

13. Known limitations (be upfront about these)

- **Integration tests use a faked database.** They prove the routes → services → repositories wiring, access control and error handling, but not Postgres itself (SQL, RLS policies and migrations are checked in Supabase and by manual UAT). A small suite against a disposable real database is the next step. Front-end browser checks are not yet a committed automated suite.
- **Single-instance assumptions.** Live-update connections, the token cache and the rate limiter are in memory. Running several server instances would need a shared store (e.g. Redis). Fine for this deployment scale.
- **Free-tier hosting** sleeps when idle; a scheduled ping to `/api/health` avoids slow first requests.
- **Optional integrations need configuration**: AI needs `ANTHROPIC_API_KEY`, email needs SMTP credentials, push needs VAPID keys. Each degrades gracefully when absent.

- **iPhone push** works only after the app is added to the Home Screen.
- Screenshots in `docs/screenshots` are generated from sample data.

14. Future work

Realtime chat between student and technician, recurring-issue detection, preventive-maintenance schedules, multi-campus support, and an SMS fallback for students without data.